Industrial Internship Report on

”Project Name”

Prepared by

[Student name]

Executive Summary
This report provides details of the Industrial Internship provided by Upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. The project was completed within a duration of 6 weeks. My project was titled **“Prediction of Agriculture Crop Yield in India”**, which involved analyzing agricultural datasets and building machine learning models to predict crop yield based on parameters such as crop type, state, and cost of cultivation. The project included: • Data preprocessing • Exploratory Data Analysis (EDA) • Model building using Linear Regression and Random Forest • Model evaluation and optimization • Deployment using Streamlit This internship provided valuable exposure to real-world industrial problems and helped in developing practical skills in data science and machine learning.

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Preface

The 6-week internship program provided an opportunity to work on a real-world industrial problem using Machine Learning and Data Analysis.

Agriculture is a critical sector in India, and predicting crop yield can help farmers and policymakers make better decisions.

During this internship:

- Worked on data preprocessing using Python
- Built machine learning models
- Learned real-world ML workflow

I would like to thank Upskill Campus, The IoT Academy, and UCT for providing this opportunity.

1 Introduction

1.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

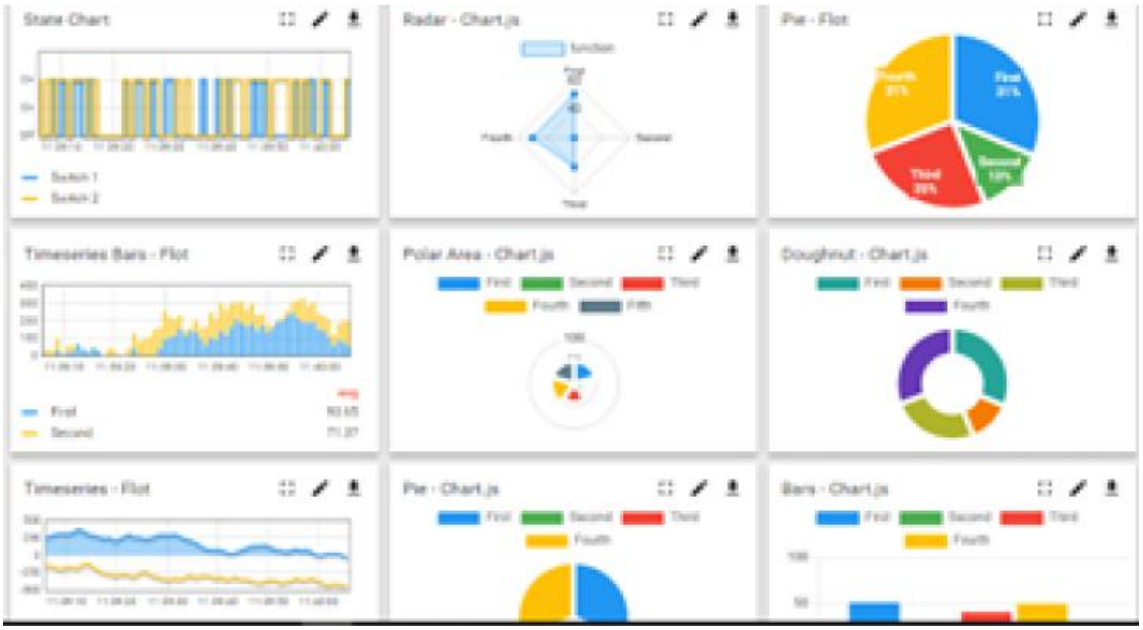
i. UCT IoT Platform (uct Insight)

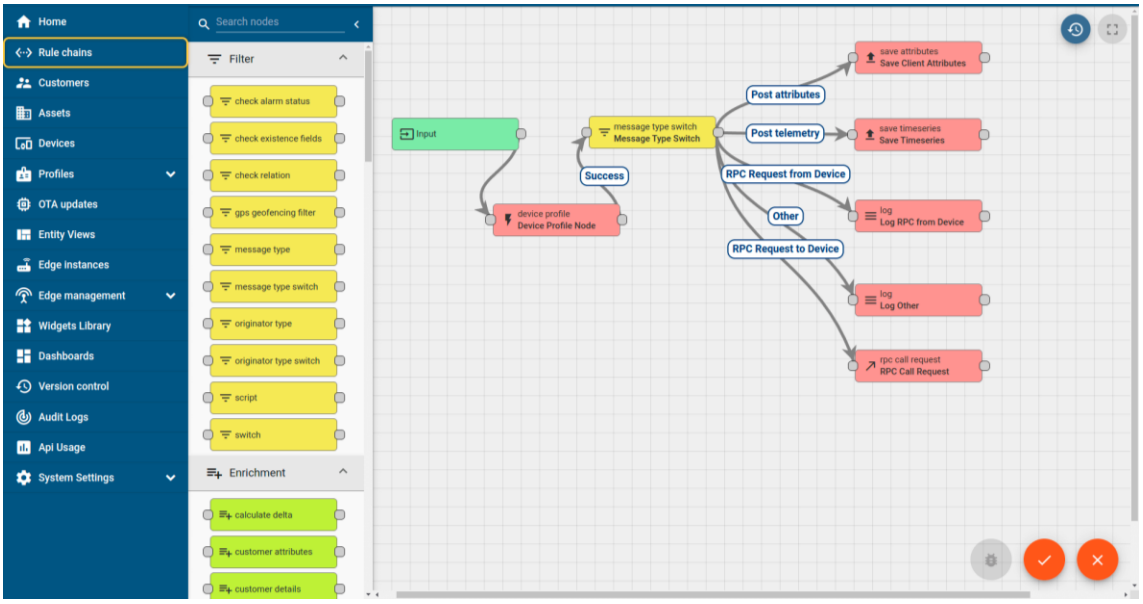
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

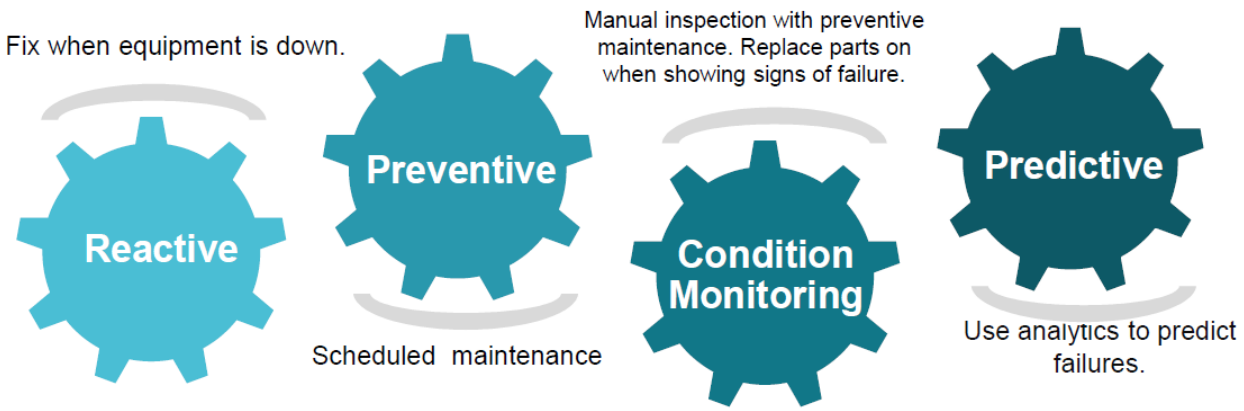
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

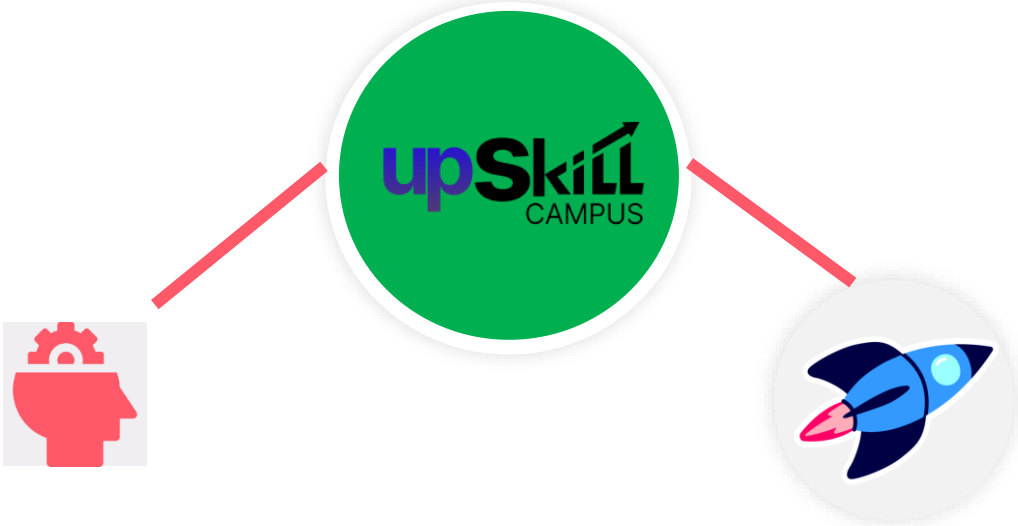
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



1.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

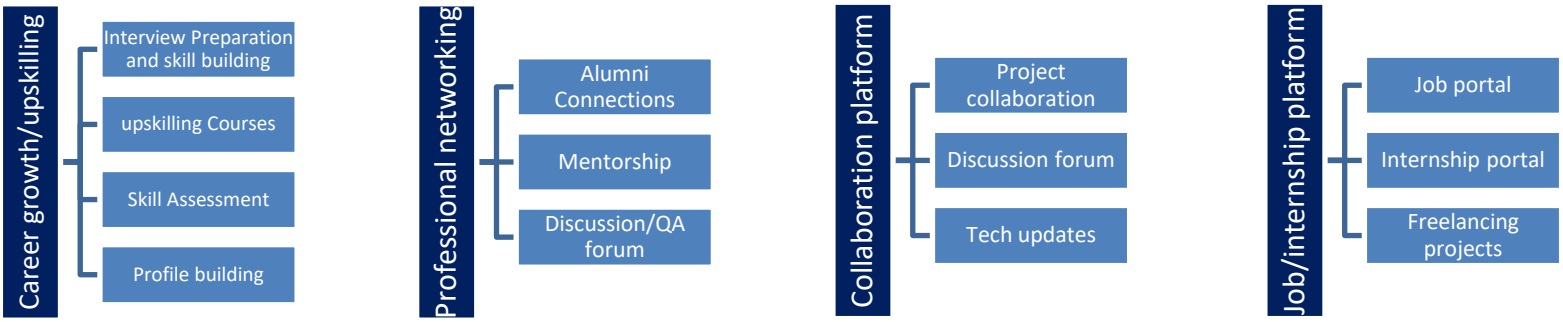
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



1.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

1.4 Objectives of this Internship program

The objective for this internship program was to

- ☛ get practical experience of working in the industry.
- ☛ to solve real world problems.
- ☛ to have improved job prospects.
- ☛ to have Improved understanding of our field and its applications.
- ☛ to have Personal growth like better communication and problem solving.

1.5 Reference

[1] Python Documentation

[2] Scikit-learn Documentation

[3] Internship Guidelines

[4] Agriculture Dataset

1.6 Glossary

Terms	Acronym
M**L**	Machine Learning
E**DA**	Exploratory Data Analysis
R^2 Score	Accuracy metric
RMSE	Error metric

2 Problem Statement

To develop a machine learning model that predicts crop yield using agricultural data such as crop type, state, and cost parameters.

Traditional methods rely on manual estimation and lack accuracy. This project aims to provide a data-driven solution.

Agriculture plays a major role in India's economy. Accurate crop production prediction is

important for:

• Farmers

• Government planning

• Supply chain management

Traditional methods rely on:

• Manual estimation

• Past experience

• Basic calculations

3 Existing and Proposed solution

Existing Solution

- Manual calculations
- Historical averages
- Low accuracy

Proposed Solution

- Machine Learning based system
- Data preprocessing and EDA
- Random Forest model

Workflow:

- Load dataset
- Clean data
- Train model
- Predict output

3.1 Code submission (Github link) :

https://github.com/jayeshkondi-oss/upskillcampus.git

3.2 Report submission (Github link) :

4 Proposed Design/ Model

The system design is represented using high-level and low-level diagrams to clearly illustrate the data flow and model workflow.

The system follows a structured pipeline:

1. Dataset Loading
2. Data Cleaning
3. Exploratory Data Analysis
4. Feature Engineering
5. Model Training
6. Model Evaluation
7. Prediction Output

4.1 Interfaces (if applicable)

The system is built using Python.

• Data is processed using Pandas

• Graphs are generated using Matplotlib

• Output is shown in console

The system is designed in a modular way and can be easily integrated into a web-based interface such as Streamlit for deployment.

5 Performance Test

Evaluation Metrics Used

- RMSE (Root Mean Square Error)
- MAE (Mean Absolute Error)
- R^2 Score

5.1 Test Plan/ Test Cases

Test Case	Input	Output
Load Data	CSV file	Dataset loaded
Cleaning	Missing data	Clean dataset
Training	Dataset Model	trained
Prediction	User input	Output generated

5.2 Test Procedure

- Loaded dataset
- Cleaned data
- Performed EDA
- Trained model
- Evaluated performance
- Tested predictions

5.3 Performance Outcome

- Random Forest achieved highest accuracy
- Reduced overfitting compared to Decision Tree
- Stable predictions across test data

6 My learnings

During this internship, I learned:

- Data preprocessing and cleaning
- Feature engineering techniques
- Implementation of ML models
- Model evaluation and tuning
- Building web apps using Streamlit
- Deploying ML projects online
- Using GitHub for version control

This experience significantly improved my confidence in working on real-world projects.

7 Future work scope

- Add more features like weather and soil data
- Improve model accuracy using advanced algorithms
- Integrate real-time data sources
- Build mobile-friendly application
- Expand to crop recommendation system